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Taking the plunge: When is best for hot water immersion to complement exercise in heat and hypoxia

Myles C. Dennis[a,b], Paul S.R. Goods[b,c,d], Martyn J. Binnie[a,b], Olivier Girard[a], Karen E. Wallman[a], Brian T. Dawson[a] and Peter Peeling[a,b]

[a]School of Human Sciences (Exercise and Sport Sciences), The University of Western Australia, WA, Australia, 6009; [b]Department of Physiology, Western Australian Institute of Sport, WA, Australia, 6010; [c]Murdoch Applied Sports Science Laboratory, Murdoch University, WA, Australia, 6150; [d]Centre for Healthy Ageing, Health Futures Institute, Murdoch University, WA, Australia, 6150

ABSTRACT

This investigation assessed the psycho-physiological and performance effects of hot water immersion (HWI) implemented either before or after a repeated-sprint training in hypoxia (RSH) session conducted in the heat. Ten participants completed three RSH trials (3 × 10 × 5-s sprints), conducted at 40°C and simulated altitude of 3000 m. A 30-min monitoring period preceded and followed all exercise sessions. In PRE, the pre-exercise period was HWI, and the post-exercise period was seated rest in temperate conditions. This combination was reversed in POST. In CON, participants were seated in temperate conditions for both periods. Compared to CON, PRE elicited a reduction in power output during each repeated-sprint set (14.8–16.2%, all $p < 0.001$), and a significantly higher core temperature (Tc) during the pre-exercise period and throughout the exercise session ($p < 0.001$ and $p = 0.025$, respectively). In POST, power output and Tc until the end of exercise were similar to CON, with Tc higher at the conclusion of the post-exercise period ($p < 0.001$). Time across the entire protocol spent ≥38.5°C Tc was significantly longer in PRE (48.1 ± 22.5 min) than POST (31.0 ± 11.3 min, $p = 0.05$) and CON (15.8 ± 16.3 min, $p < 0.001$). Employing HWI following RSH conducted in the heat provides effective outcomes regarding physiological strain and cycling performance when compared to pre-exercise or no HWI.

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KEYWORDS
Altitude; heat stress; core temperature; heat acclimation; passive heating

Introduction

Maximising physiological responses to repeated-sprint training in hypoxia (RSH), while ensuring training quality is not significantly impaired, is of interest. Combining heat with RSH may represent a potential method for maximising physiological responses; however, the impact on training quality should be considered. Prior research shows that adding heat stress during RSH can reduce constant-load cycling time to exhaustion (Girard & Racinais, 2014); however, no effect on mechanical output in fixed time efforts has been shown when compared to temperate conditions (Dennis, Goods, Binnie, Girard, Wallman, Dawson, Peeling et al., 2021; Dennis, Goods, Binnie, Girard, Wallman, Dawson, Billaut et al., 2021). Importantly, muscle deoxygenation appears augmented during the sprinting phase of RSH with the addition of heat (Dennis, Goods, Binnie, Girard, Wallman, Dawson, Billaut et al., 2021), providing an increase in physiological stimulus, since repeated fluctuations in muscular oxygen utilisation are known to improve muscle oxidative capacity (Daussin et al., 2008), and a reduced partial pressure of oxygen in muscle induces mitochondrial biogenesis (Hoppeler et al., 2003).

Interestingly, the addition of heat to an exercise session is also commonly used to achieve heat acclimation adaptations. The chronic attainment of a core temperature (Tc) ≥38.5°C represents a threshold for achieving these heat-induced adaptations, where sudomotor and vasomotor responses are challenged and heat shock proteins are expressed (Gibson et al., 2015). Recent consensus on training in the heat suggests that exercise sessions should last at least 60 min in order to produce a meaningful increase in Tc and a strong sweating response (Racinais et al., 2015). However, if we look to achieve heat adaptation whilst maximising the effect of RSH, the short-duration, "all-out" nature, of RSH protocols would make it difficult to spend long enough exposed to hot conditions for an effect, with any attempt likely increasing the overall training load on an athlete. Accordingly, alternative methods of heat exposure outside of the exercise session may need to be sought to provide a sufficient heat stimulus when combined with RSH.

Passive heat exposure is a method that can increase Tc without the imposition of an increased training load. This is commonly achieved via 30–60-min exposure to a sauna or hot water immersion (HWI) at a water temperature of ≥40°C (Heathcote et al., 2018). While heat acclimation is possible with passive heat exposure alone (Brazaitis & Skurvydas, 2010), much of the current literature has investigated passive heat exposure combined with exercise (Heathcote et al., 2018), which increases Tc due to additional heat production as an artefact of exercise-induced metabolism. Passive heat exposure has been combined with exercise sessions in both temperate and hot conditions, with physiological and performance adaptations noted such as greater plasma volume expansion,

CONTACT Myles C. Dennis myles.dennis@research.uwa.edu.au School of Human Sciences (Exercise and Sport Sciences), the University of Western Australia, Crawley, WA, Australia, 6009.

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reduced end-exercise rectal temperature and increased time to exhaustion during endurance running.(Scoon et al., 2007 and Mee and Zurawlew) et al., 2018

Investigations into the combination of HWI with exercise typically involve the HWI taking place *after* the training bout (Heathcote et al., 2018). In these scenarios, Tc has already been increased by exercise and the use of passive heat ensures sufficient exposure to an appropriate heat load (Heathcote et al., 2018). This maximises time spent under high thermal stress without the negative implication of the disruption to mechanical power output that can be experienced with a high Tc (Girard et al., 2015). However, research has also shown that effective acclimation blocks (of 5 days) can be formulated with exercise taking place after passive heat exposure (Mee et al., 2018). This work showed a 5-day heat acclimation from 20 min seated, wearing a sauna suit (50°C, 30% RH) prior to exercise resulted in a reduced Tc and heart rate (HR) at rest, as well as peak Tc, peak HR, thermal sensation (TS) and perceived exertion during a heat tolerance test (as compared to a thermoneutral control) (Mee et al., 2018). However, there is currently a lack of research that directly compares the Tc profiles of pre- *vs*. post-exercise passive heat exposure strategies on other physiological and performance consequences, and there is scant literature exploring passive heat exposure combined with RSH conducted in the heat.

Therefore, the aim of this study was to investigate the psycho-physiological and performance effects of passive heat exposure via HWI, either before or after an RSH session conducted in the heat, compared to a control with no HWI, on acute performance and physiological responses, with reference to Tc. It was hypothesised that exposure to HWI prior to the exercise session would significantly reduce repeated-sprint power output. However, it was also hypothesised that pre-exercise HWI would result in a greater duration spent with a Tc ≥38.5°C than the other conditions tested.

Methods

Participants

Ten recreationally active (McKay et al., 2022) male participants were recruited (age: 22.9 ± 3.0 y; height: 179.5 ± 7.2 cm; body mass: 76.3 ± 9.3 kg). Participants lived and trained near sea level, with no exposure to altitude (>600 m) in the month prior to testing. Trials were conducted during the autumn/winter months of Perth, Western Australia (mean monthly maximum temperature: 18.5–29.6°C) to minimise seasonal heat acclimatisation.

Written informed consent was obtained from all participants prior to commencement. Ethical approval was received from the host institution's human research ethics committee (RA/4/20/4832).

Design

We employed a randomised, repeated measures design.

Experimental overview

Participants attended one familiarisation and three experimental trials. All sessions were separated by a range of 3–14 days. During each experimental trial, participants performed a maximal exercise session in a 213 m^3 environmental chamber (The Altitude Centre, London, UK) situated near sea level (no active airflow). During all trials, the chamber was set to a constant simulated altitude of ~3000 m (F_IO_2 0.144), air temperature of 40°C and relative humidity (RH) of 50%. In all trials, the exercise session was bookended by 30-min periods of seated rest. During each of these periods, participants were either submerged to neck level in hot (40°C) water or were seated in temperate (21°C) laboratory conditions, outside the chamber. Three combinations of these conditions were tested and conducted in a randomised order determined by a random number generator:

- PRE: 30 min of HWI prior to exercise, 30 min in temperate laboratory conditions following exercise.
- POST: 30 min in temperate laboratory conditions prior to exercise, 30 min HWI following exercise.
- CON: 30 min in temperate laboratory conditions prior to and following exercise.

Participants kept a food diary for 24 h prior to each trial. During this time, they were asked to maintain a normal diet, but to avoid the consumption of alcohol. Between trials, participants continued their regular training, and the total training load in the 24 h prior to each trial was monitored and quantified as duration (min) × RPE (modified Borg CR10 scale; Christian et al., 2014). Across all entire experimental trials, participants drank water *ad libitum*. Water bottles were weighed pre- and post-trial to measure fluid intake (to 1 mL).

Experimental procedures

Familiarisation Trial: Participants initially attended the laboratory to familiarise themselves with all facilities, procedures, and equipment. This familiarisation task involved an identical exercise session to the experimental trials (see below), conducted in identical conditions (F_IO_2 0.144, 40°C, 50% RH).

Experimental Trials: Upon arrival, each testing day, participants first provided a midstream urine sample for the measurement of urine-specific gravity (U_{SG}) via a refractometer (URC-NE, 1.000-1.050, Atago, Tokyo, Japan); no participants were hypohydrated on arrival (U_{SG} > 1.030). Body mass was then measured using a digital platform scale (FG-150 K, A&D Weighing, Adelaide, Australia) to the nearest 0.05 kg. Next, participants were left in private to self-insert a rectal probe (~10 cm beyond the anal sphincter) for the measurement of Tc (Type T Thermocouple Probe RET-1, Physitemp Instruments Inc., Clifton, NJ, USA). Core temperature was continuously monitored throughout the entire testing protocol using DASYLab Light software (Version 11, National Instruments, Ireland Resources Ltd., Dublin, Ireland) sampling at 0.1 Hz.

In PRE, participants were transitioned to the water immersion area. They were submerged up to neck level in 40.1 ± 0.8°C

water, where they remained for 30 min, wearing only exercise shorts. During this time in the POST and CON participants remained seated in laboratory conditions (21.1 ± 0.8°C, 55.6 ± 9.5% RH) wearing only shorts, socks, and running shoes. Thermal sensation was assessed (9-point Likert scale; Young et al., 1987) at the start of this period, and at 5-min intervals thereafter in all conditions. At the conclusion of 30 min, participants were allowed a 2.5-min transition period to enter the environmental chamber in all conditions.

Once inside, participants in PRE first put on socks and running shoes. All conditions were then instructed to sit on a front-access air-braked cycle ergometer (Model EX-10, Repco, Melbourne, Australia). They were instrumented with a chest strap heart rate monitor (H1, Polar Electro, Kempele, Finland) and wrist-watch receiver (RS300X, Polar Electro, Kempele, Finland), as well as a wrist-worn, finger-tip pulse oximeter to measure arterial oxygen saturation (S_pO_2, WristOx2 Model 3150, Nonin Medical, Inc., Plymouth, MN, USA). Skin thermistors (Skin Sensor SST-1, Physitemp Instruments Inc., Clifton, NJ, USA) were taped to the sternum, right forearm, right mid-thigh and right mid-calf, for the measurement of skin temperature (Tsk). This procedure took no longer than 5 min. Skin temperature was monitored throughout exercise using the same software as Tc (also at 0.1 Hz). Mean Tsk was calculated using published formulae (Ramanathan, 1964).

Before exercise commencement, a capillary blood sample was taken from the earlobe and analysed for blood lactate concentration (BLa; Lactate Plus Metre, Nova Biomedical, Waltham, MA, USA). Subsequently, a 20-min standardised cycling warm-up at 120 W was completed, during which, participants performed a series of 10 × 5-s progressive sprint efforts every 2 min, with the first sprint taking place upon the commencement of the warm-up and the last sprint taking place 18 min from commencement. For each submaximal sprint, participants were instructed to work at a subjective "sense of effort" of 5, 6, 7, 7, 8, 8, 9, 9, 10 and 10 on the modified Borg CR10 "sense of effort scale" (Christian et al., 2014). Participants then commenced the exercise task, which comprised 3 × 10 × 5-s "all-out" sprints, with 20-s active recoveries (~120 W) between sprints, and an inter-set active recovery period of 2.5 min. A BLa sample was collected 60 s after the cessation of exercise, and a 10-min standardised cool-down (120 W cycling) was enforced.

Repeated-sprint ability was analysed by calculating peak power output (PPO) and mean power output (MPO) for each sprint (Cyclemax version 6.3, School of Human Sciences, UWA) averaged across each set. Additionally, total work and a percentage decrement score (from MPO) for each set were calculated (Spencer et al., 2008). Heart rate, thermal sensation and S_pO_2 were assessed before and after the warm-up, after each set of exercise, and at the conclusion of the cool-down. Rating of perceived exertion (RPE) was also recorded at each of the aforementioned time points (except for baseline) using the Borg 6–20 scale (Borg, 1982).

At the conclusion of the cool-down, participants were de-instrumented except for the Tc probe and exited the environmental chamber. Another 2.5-min transition period occurred, where participants in POST entered the water immersion area and subsequently spent 30 min submerged in hot water identically to those previously described. During this time, participants undertaking PRE and CON spent 30 min resting in temperate laboratory conditions, outside the environmental chamber. At the end of this period, the Tc probe was removed, and a post-exercise dry and semi-nude (underpants on) body mass was recorded. Whole-body sweat production was then calculated (Shapiro et al., 1982). The total protocol length for all conditions was 107.5 min.

Statistical analysis

A General Linear Mixed Model using the R (Team, 2020) package lme4 was used to analyse the data. A random intercept was included to adjust for baseline levels and inter-individual homogeneity. All models were estimated using Restricted Maximum Likelihood. Visual inspections of residual plots did not reveal any obvious deviations from homoscedasticity or normality. P-values were obtained using Type II Wald F tests with Kenward-Roger degrees of freedom as implemented in the R package car. If a significant main effect for condition or time was observed, post-hoc pairwise-comparisons using the Tukey HSD test were performed using the R package emmeans. All data are expressed as mean ± standard deviation.

Results

Significant condition effects were observed for MPO, PPO and total work done (all $p < 0.001$, Figure 1). For all three variables, PRE values were lower than those for POST and CON (all $p < 0.001$). For MPO and total work done, PRE values were 16.8% and 16.2% lower than POST and CON, respectively (all $p < 0.001$). For PPO, PRE was 16.6% lower than POST and 14.8% lower than CON (both $p < 0.001$). Significant main effects for time were also noted (all $p < 0.001$). The MPO, PPO and total work done were all higher in set 1 than both set 2 and set 3 in PRE (all $p < 0.05$). Higher MPO and total work done values were also achieved in set 1 compared to set 3 in CON (both $p = 0.029$).

There was a significant condition × time interaction for Tc measured across the entire protocol ($p < 0.001$, Table 2). After both the pre-exercise monitoring period and the exercise period, Tc values for PRE were higher than POST (both $p < 0.001$) and CON ($p < 0.001$ and $p = 0.025$ respectively); however, after the post-exercise monitoring period, POST was higher than both PRE and CON (both $p < 0.001$). Additionally, a significant condition × time interaction was observed for Tc during exercise ($p = 0.016$), demonstrating that before and after the warm-up, and after each set of repeated sprints, PRE was greater than both POST and CON (Table 1). Peak Tc was higher for both PRE (39.27 ± 0.33°C; $P = 0.01$) and POST (39.23 ± 0.39°C; $p = 0.02$) compared to CON (38.78 ± 0.36°C); PRE and POST were not different ($p = 0.955$). Total time during the protocol spent at a Tc above 38.5°C also displayed a significant condition effect ($p < 0.001$), such that PRE (48.1 ± 22.5 min) was greater than POST (31.0 ± 11.3 min; $p = 0.05$) and CON (15.8 ± 16.3 min; $p < 0.001$).

Significant condition × time interactions was observed for HR and BLa, as well as significant condition effects being noted

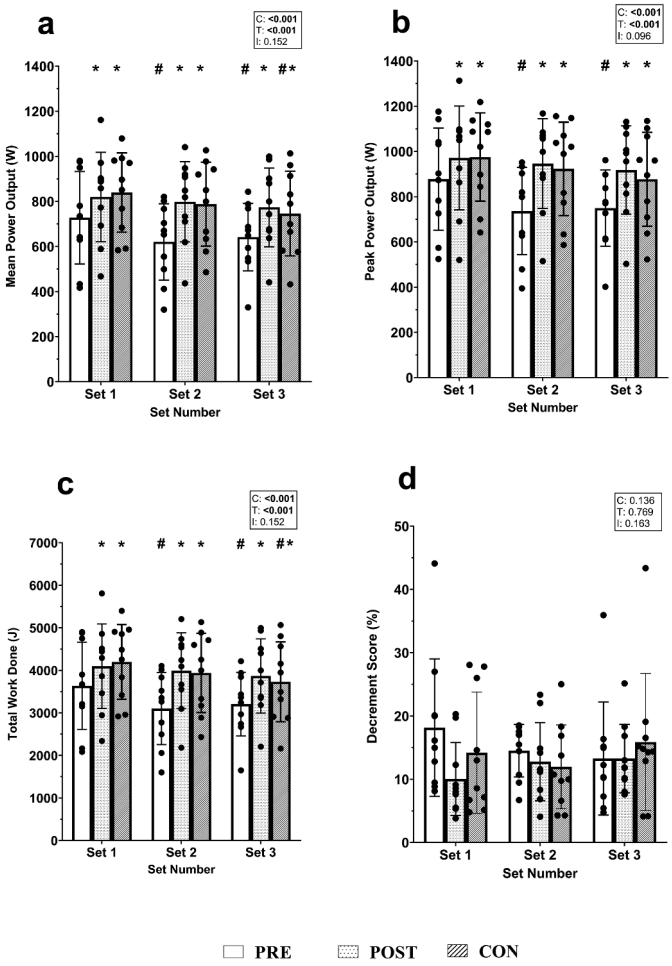

Figure 1. Mean power (a), peak power (b), total work done (c), and sprint decrement score (d) during the repeated-sprint cycling session. Note: Values are means ± SD (n = 10). The repeated-sprint session included three sets of ten 5-s sprints. Main effects for condition (C), time (T) and interaction (I) are presented. #Significant within group difference compared to Set 1. *Significant between-group difference compared to PRE.

for RPE (all $p < 0.05$). Additionally, significant time effects occurred for S_pO_2, RPE and TS (all $p < 0.05$). These data are presented in Table 3.

Participants perspired less in CON (724 ± 170 mL) compared to PRE (1767 ± 690 mL; $p < 0.001$) and POST (1613 ± 606 mL; $p < 0.001$), with no differences between PRE and POST evident ($p = 0.315$). Water intake was less in CON (955 ± 705 mL) than PRE (1357 ± 754 mL; $p < 0.001$) and POST (1493 ± 683 mL; $p < 0.001$), with no differences between PRE and POST noted ($p = 0.625$).

Discussion

The aim of the present study was to investigate the effects of passive heat exposure via HWI and determine whether it is best applied before or after RSH is conducted in the heat to maximise acute physiological responses while minimising any deleterious effect on training quality. In line with our hypotheses, we observed that exposure to passive heating prior to the exercise session significantly reduced mechanical power output (by 14-8-16.8%) across the session compared to the conditions that rested in temperate conditions. However, we also observed that this condition resulted in a significantly greater amount of time spent with a Tc above 38.5°C (48.1 ± 22.5 min) than POST (31.0 ± 11.3 min) and CON (15.8 ± 16.3 min), indicating a greater level of physiological strain while compromising power output.

The significantly reduced cycling power output experienced in PRE is likely explained by the marked increase in Tc at the onset of exercise in PRE (38.96 ± 0.49°C) compared to POST (37.23 ± 0.24°C) and CON (37.27 ± 0.42°C). This is a similar increase to previous observations utilising pre-exercise HWI where reductions in repeated-sprint performance were also found (Drust et al., 2005). Drust et al. observed that submerging participants in 41 ± 0.4°C water for 24 ± 4 min, elevating Tc to 39.6 ± 0.1°C, significantly reduced MPO and PPO during 5 × 15-s maximal sprints in hot (40.5 ± 1.0°C, 18.5 ± 1.0% RH) conditions when compared to performing 40 min of cycling at approximately 60% of VO_{2peak} in temperate (20.0 ± 1.3°C, 23.6 ± 6% RH) conditions prior to the sprints (Drust et al., 2005). Other possible explanations for decrements in repeated-sprint performance with a high Tc include mechanisms such as reductions in cardiac output impairing arterial oxygen delivery to working muscle (Sawka et al., 2011) and an inhibitory effect of the sensory afferent feedback originating from heated muscles (Nybo et al., 2014). Lower mechanical power outputs in PRE, compared

Table 1. Core and skin temperatures during the repeated-sprint cycling session (n = 10).

Core Temperature (°C)						General Linear Mixed Model		
Condition	Pre-WU	Post-WU	Set 1	Set 2	Set 3	Condition	Time	Interaction
PRE	38.96 ± 0.49	38.85 ± 0.51	38.63 ± 0.94	38.75 ± 0.65	39.02 ± 0.46	<0.001	<0.001	0.016
	(38.11 – 39.53)	(38.15 – 39.6)	(36.47 – 39.62)	(37.48 – 39.52)	(38.34 – 39.63)			
POST	37.24 ± 0.24*	37.40 ± 0.29*	37.54 ± 0.3*	37.85 ± 0.29*	38.15 ± 0.32*			
	(36.95 – 37.6)	(37.10 – 37.87)	(37.20 – 38.11)	(37.51 – 38.45)	(37.78 – 38.73)			
CON	37.27 ± 0.42*	37.59 ± 0.56*	37.77 ± 0.62*	37.92 ± 0.66*	38.12 ± 0.7*			
	(36.68 – 38.05)	(36.93 – 38.67)	(37.09 – 38.98)	(36.9 – 39.1)	(36.69 – 39.37)			
Skin Temperature (°C)								
PRE	37.23 ± 0.46	36.98 ± 0.41	36.95 ± 0.48	37.04 ± 0.59	37.25 ± 0.56	0.039	<0.001	<0.001
	(36.77 – 38.30)	(36.41 – 37.68)	(36.11 – 37.56)	(36.13 – 37.78)	(36.34 – 38.06)			
POST	35.65 ± 0.42*	36.76 ± 0.26	36.96 ± 0.3	37.23 ± 0.32	37.40 ± 0.25			
	(35.17 – 36.11)	(36.40 – 37.21)	(36.33 – 37.39)	(36.50 – 37.56)	(37.08 – 37.86)			
CON	35.66 ± 0.28*	36.76 ± 0.29	37.06 ± 0.28	37.32 ± 0.21	37.49 ± 0.22			
	(35.18 – 36.03)	(36.44 – 37.35)	(36.75 – 37.56)	(37.04 – 37.65)	(37.16 – 37.844)			

Note: Data are expressed as mean ± SD (range). *Significant between-group difference compared to PRE

Table 2. Core Temperature and Thermal Sensation responses during the entire protocol (n = 10).

Core Temperature Post-Period (°C)				General Linear Mixed Model		
Condition	Post-Monitoring Period 1	Post-Exercise	Post-Monitoring Period 2	Condition	Time	Interaction
PRE	38.71 ± 0.44	39.09 ± 0.38	37.86 ± 0.33	<0.001	<0.001	<0.001
	(37.77 – 39.22)	(38.53 – 39.77)	(37.28 – 38.28)			
POST	37.15 ± 0.25*	38.40 ± 0.28*	39.20 ± 0.40*			
	(36.83 – 37.57)	(37.87 – 38.73)	(38.60 – 39.89)			
CON	37.10 ± 0.41*	38.60 ± 0.28*	37.70 ± 0.52†			
	(36.51 – 37.87)	(38.24 – 39.03)	(37.11 – 38.75)			
Thermal Sensation Post-Period (0-8)						
PRE	7.0 ± 0.8	5.6 ± 1.6	3.9 ± 0.5	0.006	<0.001	<0.001
	(5.5 – 8.0)	(2.5 – 7.5)	(2.5 – 4.5)			
POST	3.8 ± 0.6*	5.9 ± 0.7	6.9 ± 0.9*			
	(2.5 – 4.5)	(5.0 – 7.0)	(5.0 – 8.0)			
CON	3.8 ± 0.4*	6.1 ± 0.6	4.0 ± 0.2†			
	(3.0 – 4.0)	(5.0 – 7.0)	(3.5 – 4.0)			

Note: Data are expressed as mean ± SD (range). *Significant between-group difference compared to PRE; †Significant between-group difference compared to POST.

Table 3. Physiological and perceptual responses during the repeated-sprint cycling session (n = 10). Data are expressed as mean ± SD (range).

Oxygen Saturation (%)						General Linear Mixed Model		
Condition	Pre-WU	Post-WU	Set 1	Set 2	Set 3	Condition	Time	Interaction
PRE	89 ± 2	88 ± 1	89 ± 2	89 ± 2	88 ± 2	0.118	0.002	0.141
	(85 – 92)	(87 – 89)	(84 – 92)	(86 – 92)	(85 – 90)			
POST	89 ± 2	89 ± 2	88 ± 2	87 ± 3	88 ± 2			
	(85 – 92)	(86 – 93)	(85 – 90)	(80 – 90)	(84 – 91)			
CON	89 ± 2	90 ± 2	87 ± 4	87 ± 2	87 ± 2			
	(84 – 92)	(87 – 91)	(79 – 91)	(84 – 90)	(83 – 90)			
Heart Rate (bpm)								
PRE	119 ± 12	158 ± 11	173 ± 19	173 ± 13	177 ± 13	<0.001	<0.001	<0.001
	(98 – 136)	(135 – 173)	(130 – 196)	(150 – 196)	(161 – 197)			
POST	87 ± 13*	134 ± 11*	170 ± 11	173 ± 17	177 ± 16			
	(66 – 115)	(120 – 156)	(145 – 185)	(130 – 194)	(138 – 199)			
CON	95 ± 9*	147 ± 15	175 ± 11	179 ± 10	181 ± 8			
	(83 – 106)	(131 – 175)	(161 – 189)	(168 – 194)	(170 – 197)			
Thermal Sensation (0-8)								
PRE	5.8 ± 1.1	5.9 ± 1.2	6.5 ± 1.4	6.9 ± 1.0[ab]	6.8 ± 1.4[a]	0.334	<0.001	0.656
	(4.0 – 7.5)	(4 – 7.5)	(3.0 – 8.0)	(4.5 – 8.0)	(3.0 – 8.0)			
POST	5.3 ± 0.3	5.8 ± 0.6	6.4 ± 0.8[a]	6.9 ± 0.7[ab]	7.1 ± 0.6[ab]			
	(5.0 – 6.0)	(5.0 – 7.0)	(5.0 – 7.5)	(5.5 – 7.5)	(6.0 – 8.0)			
CON	5.5 ± 0.5	6.1 ± 0.6	6.5 ± 0.5[a]	7.1 ± 0.4[a]	7.2 ± 0.3[ab]			
	(5.0 – 6.0)	(5.5 – 7.0)	(6.0 – 7.0)	(6.5 – 7.5)	(6.5 – 7.5)			
Rating of Perceived Exertion (6-20)								
PRE	-	14 ± 2	18 ± 1[b]	18 ± 2[b]	19 ± 1[b]	<0.001	<0.001	0.114
		(11 – 17)	(15 – 19)	(14 – 20)	(16 – 20)			
POST	-	13 ± 2*	16 ± 2*[b]	18 ± 2*[bc]	19 ± 1*[bc]			
		(11 – 16)	(13 – 18)	(14 – 19)	(15 – 20)			
CON	-	13 ± 1†	17 ± 1†[b]	18 ± 1†[bc]	19 ± 1†[bc]			
		(11 – 16)	(15 – 18)	(17 – 19)	(18 – 20)			
Blood Lactate (mmol/L)								
PRE	1.9 ± 0.6	-	-	-	9.8 ± 2.3	0.033	<0.001	0.003
	(1.1 – 2.8)				(4.0 – 13.0)			
POST	1.6 ± 0.9	-	-	-	11.9 ± 1.8*			
	(0.4 – 2.8)				(8.3 – 14)			
CON	1.7 ± 0.8	-	-	-	12.5 ± 2.7*			
	(0.7 – 2.8)				(7.7 – 16)			

Note: [a]Significant within-group difference compared to pre-warm-up; [b]Significant within-group difference compared to post-warm-up; [c]Significant within-group difference compared to Set 1; *Significant between-group difference compared to PRE. †Significant between-group difference compared to POST.

to POST and CON (unaffected by increases in Tc), present a weaker training stimulus resulting from the combined heat and RSH session. As such, if the aim of a practitioner is to prescribe a combined heat and RSH session with the goal of optimising power output, then it is inappropriate to utilise HWI preceding exercise.

Despite a reduced power output being observed in PRE compared to the other conditions, PRE also seemingly displayed the greatest heat stress dose across the entirety of the protocol. Compared to PRE, where total time spent with Tc ≥38.5°C was ~45 min, approximately two-thirds of this stimulus (~30 min) was observed in POST, and approximately one-third (~15 min) in CON. Of note, these differences, as well as the lack of differences in peak Tc in the current study between PRE and POST, which were both greater than CON, indicate that HWI was effective in elevating Tc regardless of timing. A Tc of 38.5°C represents a threshold above which heat acclimation-related adaptations are more likely to occur (Garrett et al., 2009). Importantly, however, participants undertaking POST finished the testing protocol with a mean Tc of 39.20 ± 0.40°C, indicating that there was some time between

the conclusion of the protocol and the time at which the body would have undergone thermoregulation to restore Tc to <38.5°C. Our data showed that, during PRE, participants took approximately 15 min to cool from a similar Tc, achieved after the conclusion of the exercise period (39.09 ± 0.38°C), to 38.5°C, during seated rest in temperate conditions. Therefore, we could reasonably extrapolate this to the time period after the conclusion of the protocol for POST, and conclude that it is likely that the actual time spent with Tc ≥38.5°C during POST was approximately 45 min (although this requires confirmation and is a limitation of the current investigation). This indicates that overall, POST likely provided a similar heat stress dose to PRE. There is also a strong evidence base to suggest that post-exercise HWI can be an effective method for inducing heat acclimation, even if exercise is performed in temperate conditions. Studies published by Zurawlew et al. investigated 6 days employing 40 min of HWI following 40 min of running at ~65% of VO_{2max} in temperate conditions (20°C) and demonstrated physiological and perceptual heat acclimation benefits (Zurawlew et al., 2018, 2019). Additionally, it has been found that heat acclimation induced via post-exercise HWI can elicit greater thermal adaptations than when induced by exercise alone (McIntyre et al., 2021). As such, when implementing the POST protocol for the purpose of maximising both mechanical power output and physiological strain, it is likely that it provides a sufficient physiological stimulus for a heat acclimation response if applied appropriately.

Although power output was lower, greater RPE values during PRE compared to POST and CON indicated that participants exerted themselves significantly more to produce this suppressed power output. It is likely that this relates to the hyperthermic Tc being experienced during PRE. This is a similar effect to what has been observed previously by Gregson et al. when implementing pre-warming before exercise in hot conditions (Gregson et al., 2002). These researchers attributed this effect to the increased contribution of the central nervous system to fatigue development, which is not present at a lower Tc (Pitsiladis et al., 2002). Furthermore, TS values were no different between conditions, meaning that despite a higher Tc during PRE, this did not translate into a difference in thermal perception. Similarly, Dennis et al., in a similar RSH session to the current study, found that Tc did not differ between conditions, but TS was significantly greater at 35°C and 40°C than at 20°C (Dennis, Goods, Binnie, Girard, Wallman, Dawson, Billaut et al., 2021). Additionally, Tsk was greater in 40°C compared to 35°C, which in turn was greater than 20°C, whereas in the current study Tsk was not different between conditions from any point during exercise after the warm-up. As such, it may be concluded that ambient air temperature and Tsk may be more closely linked to thermal perception than Tc during RSH (Dennis, Goods, Binnie, Girard, Wallman, Dawson, Billaut et al., 2021). Overall, these measures indicate that POST is likely the most suitable HWI protocol to implement in a high-performance setting. This is exhibited by lower RPE values while maintaining an increased mechanical output and constant TS, with this being more likely to be well tolerated as part of a periodised training programme.

Additionally, during exercise, HR did not differ between conditions occurring despite a lower power output in PRE compared to POST and CON. An increased Tc places a higher thermoregulatory demand on the body, eliciting a subsequent increase in HR to enable additional blood flow to the periphery for thermoregulation purposes (Casa, 1999). Finally, the increase in BLa from baseline to the end of exercise was smaller in PRE compared to the other conditions, likely a side effect of reduced mechanical power output. This may indicate an absolute reduction in glycolytic activity during PRE compared to POST and CON, which is an indirect by-product of an increased thermal strain.

Limitations and future research

Because we assessed the acute effects of passive heat exposure on psycho-physiological and performance responses to combined heat and RSH exercise, the chronic effects of multiple exposures to this type of session could not yet be determined. While POST did not display any negative effects compared to CON in terms of key session outcomes, none of our observations would encourage the use of POST over CON in an acute setting. While literature findings suggest that chronic exposure to POST would elicit both heat acclimation (Zurawlew et al., 2018; evidenced by increased thermal strain) and RSH-based benefits (Hoppeler et al., 2003), any comment on the efficacy of this type of exposure remains speculative. As such, future research investigating the effect of chronically applying POST and CON protocols is critical before implementation by practitioners. In addition, our experimental design did not allow for measurement of Tc after the conclusion of HWI in POST. If Tc measurement was obtained during the period after completion of HWI, then this would have allowed for further interpretation of Tc data. Finally, although participants were categorized based on a recently published participant classification framework (McKay et al., 2022), specific measures of participant fitness (e.g., maximal oxygen consumption) were not included. There is some evidence to suggest that the level of physical fitness can influence the effectiveness of heat acclimation (Pandolf et al., 1977), and as such may need to be considered when interpreting the final outcomes.

Conclusion

Our results show that employing HWI following RSH completed in hot ambient conditions elicits a more effective cycling performance when compared to conducting HWI prior to exercise, and a greater thermal strain than not conducting HWI. This type of session does not seem to provoke exacerbated perceptions of fatigue or increases in cardiovascular strain. Overall, this protocol is suitable for implementation in a high-performance setting, and the results can be used to guide practitioners when trying to elicit the greatest possible benefit from combining passive heating with RSH conducted in hot ambient conditions.

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ORCID

Myles C. Dennis http://orcid.org/0000-0002-9756-3584

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